

REVIEW



A critical review of the neurovascular nature of migraine and the main mechanisms of action of prophylactic antimigraine medications

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ABSTRACT

Introduction: Migraine involves neurovascular, functional, and anatomical alterations. Migraineurs experience an intense unilateral and pulsatile headache frequently accompanied with vomiting, nausea, photophobia, etc. Although there is no ideal preventive medication, frequency in migraine days may be partially decreased by some prophylactics, including antihypertensives, antidepressants, antiepileptics, and CGRPergic inhibitors. However, the mechanisms of action involved in antimigraine prophylaxis remain elusive.

Areas covered: This review recaps some of the main neurovascular phenomena related to migraine and currently available preventive medications. Moreover, it discusses the major mechanisms of action of the recommended prophylactic medications.

Expert opinion: In the last three years, migraine prophylaxis has evolved from nonspecific to specific antimigraine treatments. Overall, nonspecific treatments mainly involve neural actions, whereas specific pharmacotherapy (represented by CGRP receptor antagonists and CGRPergic monoclonal antibodies) is predominantly mediated by neurovascular mechanisms that may include, among others: (i) reduction in the cortical spreading depression (CSD)-associated events; (ii) inhibition of pain sensitization; (iii) blockade of neurogenic inflammation; and/or (iv) increase in cranial vascular tone. Accordingly, the novel antimigraine prophylaxis promises to be more effective, devoid of significant adverse effects (unlike nonspecific treatments), and more beneficial for the quality of life of migraineurs.

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1. Introduction

Migraine is a group of primary headache disorders, which are mostly unilateral, throbbing and long-lasting; prone to worsen with physical activity; and sometimes include photophobia/phonophobia and/or nausea/vomiting [1]. This neurovascular disorder may be phenomenologically classified into two major types: (i) migraine with aura, when the headache is preceded by a compendium of neurological disturbances (mainly visual); and (ii) migraine without aura, with headache as the main manifestation [1]. Moreover, other prominent features of migraine include: (i) its episodic (<14 headache days per month) or chronic (>15 headache days monthly for three months) [1,2] nature; (ii) its potential association with suicide ideations/attempts [3]; (iii) its predisposition for developing comorbidities such as insomnia, anxiety, depression, etc. [4–9]; and (iv) its association with miscellaneous, functional, molecular, genetic, and anatomical neurovascular alterations, which may increase the frequency of attacks [4–9]. Interestingly, migraine is 2–3 times more frequent in women, which may be related with the presence of estrogenic receptors in trigeminal and CGRP-containing neurons [10].

Although the history of migraine and its treatment spans millennia, the last four decades have witnessed a significant progress in identifying the pathophysiological mechanisms involved in migraine and in the development of specific antimigraine drugs [11].

Accordingly, antimigraine pharmacological treatment can be classified into acute (abortive) and prophylactic (preventive) antimigraine drugs [11], and both pharmacotherapies can be specific (i.e. ergots, triptans, ditans, gepants, CGRPergic monoclonal antibodies [mAbs]) or nonspecific (i.e. nonsteroidal anti-inflammatory drugs [NSAIDs], antiepileptics, β -blockers, and antidepressants) [12]. While most patients need acute treatment, recurrent migraine attacks (in which abortive therapy fails or is overused) may require prophylactic treatment [13]. Hence, prophylactic treatment is mainly focused on: (i) the reduction of attack's frequency and/or severity; and (ii) the improvement of drug responsiveness to acute attacks [13].

In general, the mechanisms of action that explain the efficacy of prophylactic antimigraine medications remain mainly unknown or there is controversy on their sites of action (i.e. neural vs. vascular mechanisms [11,12]). On this basis, the aim of the present review was to analyze and discuss: (i) the

Article highlights

- Migraine is a complex neurovascular disorder that involves calcitonin gene-related peptide (CGRP) released from sensory perivascular nerves following activation of the trigeminovascular system.
- Current antimigraine pharmacotherapy can be divided into acute (abortive) and prophylactic (preventive) treatments.
- Furthermore, antimigraine drugs can be classified into specific and nonspecific. Specific drugs mainly block (directly or indirectly) the CGRPergic transmission at the peripheral and central levels, whereas nonspecific medications refer to molecules whose target was not directly related to antimigraine treatment.
- Although triptans (specific acute drugs) are the gold standard in antimigraine therapy, some patients with severe and/or high-frequency attacks require prophylactic (specific or nonspecific) treatment to reduce the frequency and severity of attacks.
- The nonspecific prophylactic treatments (i.e. some antiepileptics, antihypertensives [β -blockers], and antidepressants) mainly target the neuronal excitability of the central nervous system. However, they also seem to exert a minor indirect vascular action.
- In contrast, specific prophylactic treatments (i.e. the CGRP receptor antagonist atogepant and CGRPergic monoclonal antibodies) as well as botulinum toxin type A (injected pericranially) directly interfere with the CGRPergic transmission at the peripheral (neurovascular), rather than central, levels. In addition, some second generation (longer-acting) triptans (i.e. frovatriptan, naratriptan and zolmitriptan) have also shown efficacy in the short-term prophylactic treatment against menstrual migraine.
- In general, only the specific prophylactic treatments seem to interact with vascular function by: (i) blocking vascular CGRP receptors (e.g. atogepant and the mAb erenumab); (ii) scavenging CGRP (e.g. mAbs against CGRP including galcanezumab, fremanezumab, and eptinezumab); (iii) interfering with CGRP release (e.g. prejunctional trigeminal inhibition by the triptans); and/or (iv) producing cranial vasoconstriction (e.g. via 5-HT_{1B} receptors by the triptans). However, the novel prophylactic antimigraine treatments (atogepant and anti-CGRPergic mAbs) mainly involve the first two mechanisms. These novel treatments promise to be more effective and beneficial to the quality of life of migraine patients.

complex neurovascular nature of migraine; and (ii) the mechanisms that may underlie the pharmacological actions of the main prophylactic antimigraine groups (including antiepileptics, β -blockers, long-acting triptans, antidepressants, and CGRPergic mAbs).

2. The neurovascular nature of migraine: beyond vasodilatation

As previously documented in other reviews dealing with migraine pathophysiology [11–19]: (i) migraine is a neurovascular disorder characterized by the key involvement of the trigeminovascular system; and (ii) its pathophysiology may involve, at least, two critical events in plasma (resulting in cranial vasodilatation), namely, low levels of circulating serotonin (5-hydroxytryptamine; 5-HT) and/or high levels of circulating CGRP. Undoubtedly, these advances have led to important therapeutic breakthroughs in acute and prophylactic antimigraine treatment [11–19].

Acute and prophylactic pharmacotherapies display striking differences in their mechanisms of action, efficacy, and adverse side effects [12]. From a historical perspective, the specific pharmacological treatment of migraine started with the use of ergots (potent vasoconstrictors) and, subsequently (in the 90's), with the development of the triptans (selective

cranial vasoconstrictors) [11]. Certainly, current specific pharmacotherapy has tried to avoid vasoconstrictor effects, which may cause serious adverse events in patients with cardiovascular risks [19].

Admittedly, in addition to CGRP, other neuropeptides seem to be involved in migraine pathophysiology and include, among others: (i) pituitary adenylate cyclase-activating polypeptide (PACAP); (ii) vasoactive intestinal peptide (VIP); (iii) neuropeptide Y (NPY); (iv) substance P (SP); and (v) orexins [16,20–22].

Notably, the neurological aspects of migraine (e.g. a thalamocortical dysrhythmia [23]) have recently been investigated and gained weight, placing the vascular mechanisms on the back burner [24] (Figure 1). Nevertheless, the neurovascular interaction goes beyond the simple vasocontractile/vasorelaxant control [25] (Figure 1). In fact, vascular and neural interactions: (i) coexist in a perpetual bidirectional exchange that controls the lumen of brain and cranial blood vessels [26,27]; and (ii) involve the integration and maintenance of a highly specialized blood–brain barrier (BBB) whose disruption may produce metabolic alterations in the neural environment [28]. Moreover, a diversity of brain supply arteries and anastomoses (e.g. the circle of Willis) guarantee the supply of metabolic precursors, the electrolytic homeostasis, and the hormonal communication brain/rest of the body [8,29]. At the same time, in some vascular beds, there is a 'neurovascular congruence' that can be identified by perivascular nerve–blood vessel alignments throughout the body (e.g. the carotid artery and the vagus nerve) which allow a local instantaneous tissue-specific modulation [30]. Hence, neural activity cannot be understood in isolation from vascular activity. The vascular contribution to migraine's pathophysiology may be direct (e.g. vascular aberrations, such as an incomplete circle of Willis [8,9]) and indirect via the perpetual neurovascular bidirectional interactions (see sections 2.1. and 2.2.).

Interestingly, the efficacy of non-contractile antimigraine agents [49] has been used as an argument to propose migraine as an entirely neurogenic pathology [50]. Nevertheless, those agents lacking contractile actions in mammal arteries (e.g. lasmiditan) still prevent trigeminal CGRP release [51] and, hence, both its direct vascular (i.e. vasodilatation) and neural (i.e. pain transmission) actions. On the other hand, not every section of the intracranial blood vessels seems to be phenomenologically associated with migraine, and some studies have omitted this peculiarity, as examined recently by Prof. Shevel [52]. In support of the above notion, cautery ablation of selected terminal branches of the external carotid artery was reported to suppress sporadic hemiplegic migraine in three patients [53], suggesting a critical vascular role of these terminal branches in these specific cases. In fact, an interesting method for detecting when migraines have a cranial vascular origin of pain has been recently proposed by Prof. Shevel [54], suggesting that cranial blood vessels are also essential elements in the pathophysiology of migraine.

To sum up, as previously established [1,2], migraine is a multisystem pathology that includes different types (e.g. migraine with typical aura, migraine with brainstem aura,

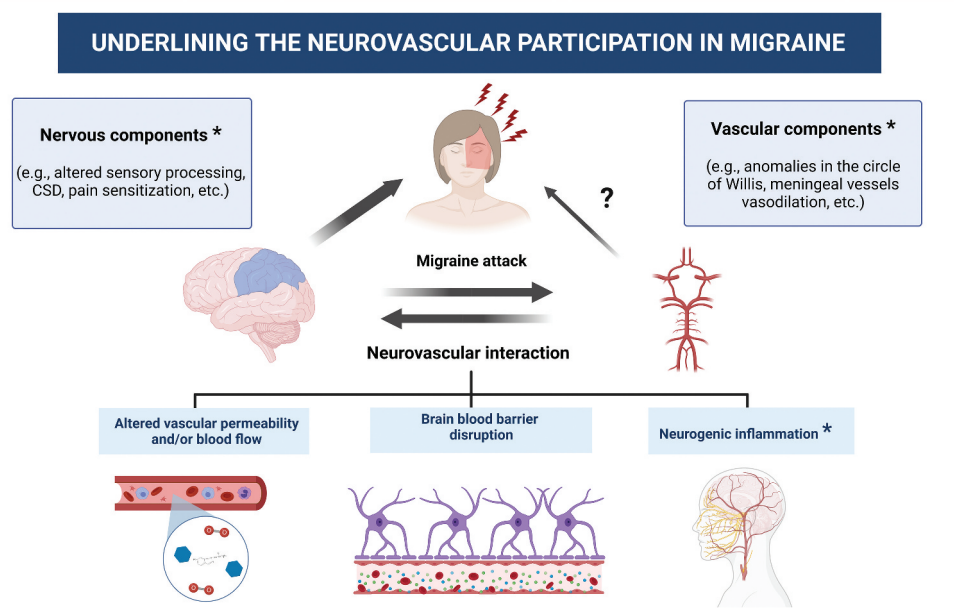


Figure 1. Major aspects of the neurovascular axis interaction and its impact on the pathophysiology of migraine.

Notes: (i) the nervous components seem to explain mainly the symptomatology of migraine, whereas the vascular components may contribute directly (e.g. circle of Willis anomalies) and indirectly via the neurovascular interaction (e.g. by promoting a peripheral sensitization); (ii) neurogenic inflammation consists of neurogenic vasodilatation (NV; mediated by CGRP release) and plasma protein extravasation (PPE; mediated by SP, NKA, and ET3) [121,122]; and (iii) NV, but not PPE, seems to be involved in migraine pathophysiology (see text for further details [121,122]). * Indicates potential sites of action (resulting in inhibition/blockade) for prophylactic antimigraine drugs, as further summarized in Table 1

Table 1. Preclinical and clinical evidence of the common neurovascular effects of prophylactic (preventive) antimigraine medications.

Pharmacological Group	Compounds	Prevention of CSD -associated events	Inhibition of Pain Sensitization	Blockade of Neurogenic Inflammation**	References
Antiepileptics	Topiramate	☑	☑	☑	[31,145]
	Sodium valproate/ divalproex sodium	☑	☑	☑	[32–34]
β-Blockers	Propranolol	☑	☑	☑	[35,36,116]
	Metoprolol	☑	X	X	[37–40]
	Timolol	N/A	N/A	X	[41]
	Atenolol	N/A*	☑	N/A	[42,173]
Triptans	Naratriptan	N/A	☑	☑	[43,44]
	Zolmitriptan	N/A	☑	☑	[44,45]
Antidepressants	Venlafaxine	N/A	☑	☑	[46,178]
	Amitriptyline	☑	☑	☑	[47,48,103]

Abbreviations: CSD, cortical-spreading depression; CNS, central nervous system; N/A, not available information; X, not reached effect; N/A*, atenolol penetration into the CNS is very low; thus, no CSD prevention should be expected. ** Neurogenic inflammation consists of plasma protein extravasation (PPE; mediated by SP, NKA, and ET3) and neurogenic vasodilatation (NV; mediated by CGRP release); however, NV, but not PPE, seems to be involved in migraine pathophysiology (see text for further details [121,122]).

hemiplegic migraine, migraine without aura, etc.) and, thus, there is no general pathophysiological explanation for all cases. Moreover, the vascular interaction with the nervous system goes beyond the simple direct control of the vascular lumen. For example, the participation of the BBB and aberrations in brain vascular supply structures (e.g. the circle of Willis) require further investigation in the pathophysiology of migraine as they may produce dysregulation in the neural environment and the cerebral permeability and blood flow, respectively (see below).

2.1. The blood-brain barrier (BBB)

The BBB regulates the permeability of substances to the central nervous system (CNS) [55]. The cells that make up the BBB interact with each other, but also with circulating cells (i.e. red

and white blood cells) and with perivascular sympathetic and sensory nerve terminals via the release of neurotransmitters, hormones, autacoids, lipids, and cytokines [56–59]. Just to provide an example of this interaction, endothelial cells may modulate the sympathetic release of noradrenaline [60]. Evidently, the penetration of molecules into the CNS involves additional mechanisms such as membrane transporters (e.g. P-glycoprotein, P-gp; ATP-binding cassette, ABC) [61], which also seem to participate in different pathologies and/or therapies, including migraine [62,63].

Disruption of the BBB results in extravasation of molecules, but it also produces alterations in the electrolytic concentrations in the cerebrospinal fluid (CSF) [64–66]. Hence, the BBB and its interactions form a neurovascular structure that regulates the CNS biochemical homeostasis [59]. Indeed, in migraine patients: (i) brain vasculature changes seem to affect BBB integrity which, in turn, leads to plasma protein

extravasation, neuroimmune inflammation and meningeal electrolytic alterations [67–69]; (ii) BBB permeability changes alter the biochemical environment of meningeal trigeminovascular nociceptors [70], producing facial hyperalgesia and other head-tissues pains [71,72]; and (iii) there are other functional and structural brain changes that may promote a sensory hyperexcitability (e.g. alterations in gray matter density and cortical and subcortical neurochemistry) [70,73–75]. Interestingly, a study in 19 patients with migraine with aura showed a normal permeability of the BBB during spontaneous migraine attacks [76], while another study reported BBB alterations in the migraine population [77]; hence, future research should allow a clear view of the BBB status in migraine. However, higher albumin and fibrinogen levels were detected in the CSF samples from 42 patients with episodic and chronic migraine vs. 14 healthy volunteers [78]; this study suggests that, at least, blood–CSF barrier transport may be altered in migraine. Despite these controversial findings, telcagepant (a CGRP receptor antagonist) achieves low CGRP occupation in the CNS [79], suggesting that its abortive antimigraine effects could be placed out of the CNS [80]. Accordingly, BBB alterations in migraine remain controversial [81].

2.2. Functional and anatomical aberrations in brain vascular supply arteries

It is noteworthy that cortical surface blood vessels are under the control of trigeminal sensory and autonomic neurons (from the superior cervical ganglion [sympathetic rami]) and the sphenopalatine and otic ganglia (parasympathetic rami) [82,83]. Hence, hormones, peptides, and neuromodulators from sensory neurons, autonomic neurons, and astrocytes, among others (e.g. endothelial cells), continuously regulate the vascular diameter [82,84]. As the catecholaminergic sympathetic division controls vascular tone [85], monoamine receptors exert a key modulation of sensory and autonomic perivascular activities [86].

Moreover, it is to be highlighted that: (i) the circle of Willis is important for the brain blood supply, the exchange between the anterior and the posterior circulation, and the brain hemispheres [29]; (ii) migraine susceptibility may be associated with abnormalities in blood vessel structures, such as the circle of Willis; and (iii) infarcts in the posterior circulation territory are ten-fold higher in patients with migraines with aura than in non-migraine patients, and three-fold higher than in patients without aura [87]. Hence, there is a potential association between vascular aberrations (e.g. the circle of Willis) and migraine (especially with aura).

Furthermore, there is a neurovascular functional congruence (i.e. a blood flow-brain activity axis) that could be altered during migraine with aura, namely ‘cortical spreading depression’ (CSD). The CSD is a wave of electrophysiological hyperactivity followed by a wave of inhibition; generally, it consists of a slow propagating (around 3 mm/min) wave of neuronal depolarization associated to migraine with aura [88]. Interestingly, acute sleep deprivation enhanced the experimentally induced CSD in rodents [89], which may explain the association between sleep alterations and migraine. This CSD involves a transitory increase in blood flow followed by a long-

lasting decrease [90], suggesting a solid functional relationship (i.e. congruence) between blood vessels and nervous activity [88,91]. In fact, after CSD, there is a loss in vascular reactivity, which may be seen as sloppy in the functional vascular/nervous congruence [90]. Moreover: (i) the effectivity of cervical noninvasive vagal stimulation against migraine has recently been confirmed by a meta-analysis [92], so that the vagal nerve may be involved in migraine headache [93]; and (ii) the greater occipital nerve may regulate sensory responses in the trigeminocervical complex by GABAergic and glycinergic mechanisms [94], suggesting that other non-pharmacological alternatives to treat and prevent migraine might be achieved.

In summary, pathologies that involve changes in brain activity (e.g. migraine) and/or a direct or indirect role of the cardiovascular system are not mutually exclusive, as changes in nervous activity are accompanied by changes in cardiovascular function and vice versa.

3. Neurovascular actions of the recommended prophylactic antimigraine medications

Preclinically, several drugs show the potential to produce prophylactic (preventive) antimigraine actions. Nevertheless, according to the last evidence-based guide for clinicians from the American Academy of Neurology [95], the prophylactic medications with strong (level A) to moderate (level B) efficacy to prevent migraine attacks (considered in this review), are included in the following groups: (i) the antiepileptics topiramate, divalproex sodium, and sodium valproate; (ii) the β -blockers propranolol, metoprolol, timolol, and atenolol; (iii) the long-acting triptans frovatriptan, naratriptan, and zolmitriptan; and (iv) the antidepressants venlafaxine and amitriptyline (see Table 1 and Figure 2). Notably, the above medications (except for the triptans) are classified as nonspecific antimigraine treatments, and all these molecules exert an action on neuronal excitability; this action highlights the relevance of neural mechanisms in prophylactic antimigraine treatment [12]. Furthermore, although not discussed in this section, monoclonal antibodies (mAbs) against CGRPergic transmission as well as some gepants (CGRP receptor antagonists) have recently emerged as specific prophylactic antimigraine medications [17] (see Figure 2).

The antimigraine activity of other complementary substances that include NSAIDs, vitamins, nutraceuticals, etc., has also been investigated [96–98]. However, they have not been included in this review as the available information requires an independent analysis.

Remarkably, prophylactic medications are underutilized, as only 13% of the possible migraine patients could benefit from them [99]. Moreover, the specific mechanisms of action of most prophylactic drugs remain mainly unknown/unproven, but some phenomenological lines of evidence [100,101] support at least four common pathways, namely: (i) a reduction in the CSD-associated events; (ii) an inhibition of central and peripheral sensitization of pain pathways; (iii) a blockade of neurogenic

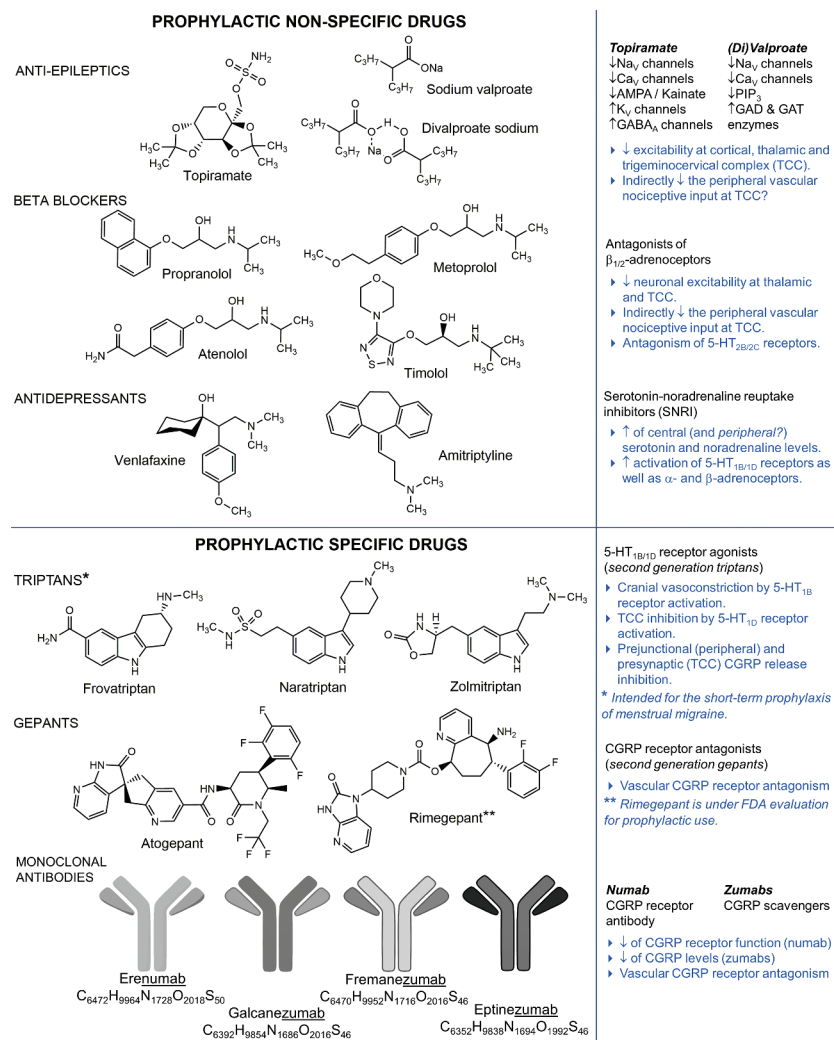


Figure 2. Specific and nonspecific prophylactic antimigraine drugs and their neurovascular actions.

*Notes: These triptans can also interact with 5-HT_{1F} receptors (pK_i values between 7.0 and 8.7 [11]).

inflammation; and (iv) an increase in cranial vascular tone and other vascular actions [11,19,101].

3.1. A reduction in the CSD-associated events

Some frequent symptoms of migraine (especially migraines with aura) such as trigeminal sensitization, photophobia, and reduced locomotion are related to the CSD phenomenon [102]. Interestingly, acute (abortive) antimigraine drugs such as sumatriptan (a 5-HT_{1B/1D/1F} receptor agonist) and olcegepant (a CGRP receptor antagonist) stopped those CSD-related events [102]. Moreover, chronic (but not acute) topiramate, valproate, amitriptyline, propranolol, and methysergide reduced the frequency of preclinically evoked CSD [103]. Similar results were found with flunarizine [104], lamotrigine [105], and the CGRP receptor antagonists MK-8825 and olcegepant [106]. Thus, a potential mechanism of action for some prophylactic medications may be their efficacy for preventing CSD or for decreasing its associated events.

3.2. An inhibition of central and peripheral sensitization of pain pathways

Central and peripheral sensitization are experienced by migraine patients, in which physical stimuli generate amplified responses that may produce allodynia and hyperalgesia [107,108]. Indeed, neuroimaging studies suggest that migraine patients have altered brain functions in areas involved in pain integration, such as the periaqueductal gray [109], insula [110], and amygdala [111]. The alterations in pain-related brain areas in migraine may lead to deficiency of the descending pain pathways (i.e. losing pain inhibition capacity) and/or hyperexcitability of nociceptive pathways [109–111]. For example, botulinum toxin type A (BOTOX; a prophylactic agent) inhibits peripheral and central sensitization in chronic migraine by inhibiting glutamate release [112,113] or modulates inflammatory mediators such as the inducible nitric oxide synthase (iNOS) and interleukin-1β in the trigeminal ganglion [114], among other effects (see below). On the other hand, serotonergic and noradrenergic reuptake inhibitors (SNRI), which

may induce preventive antimigraine actions, enhance the activity of descending pain-modulating pathways that project to the spinal trigeminal nucleus and inhibit central sensitization [115]. Similarly: (i) propranolol exerts its preventive action, at least in part, via blocking the chronic sensitization of descending controls of pain [116]; and (ii) fremanezumab, a humanized monoclonal anti-CGRP antibody, also prevents the activation and sensitization of high threshold trigemino-vascular neurons induced by CSD [117]. Thus, inhibition of peripheral and central sensitization represents a preventive antimigraine mechanism of action.

3.3. A blockade of neurogenic inflammation

Overall, the release of proinflammatory mediators, including tumor necrosis factor α (TNF- α), interleukin-1 β (IL-1 β) and interleukin-6 (IL-6) may be involved in the pathogenesis of migraine [118,119]. Therefore, the antimigraine effects of some preventive drugs may also be due to inhibition of the release of these mediators or their direct effects. In this regard, Hirfanoglu et al. [120] showed that the levels of TNF- α , IL-1 β , and IL-6 were significantly lower after four months of therapy with cyproheptadine, amitriptyline, propranolol, or flunarizine. Nonetheless, no differences were found for cytokine diminution of patients with a 50% reduction in headaches compared with those of patients with no reduction. Despite these findings, Williamson & Hargreaves [121] and Peroutka [122] have proposed that neurogenic inflammation (NI), which is induced by trigeminal release of neuropeptides, tachykinins and other mediators, consists of two separate physiological components, namely: (i) neurogenic vasodilatation (NV), primarily mediated by CGRP release at the neurovascular level; and (ii) plasma protein extravasation (PPE), mediated by tachykinins (substance P [SP] and neurokinin A [NKA]) and endothelin-3 (ET3) [121,122].

On this basis, specific antimigraine drugs (e.g. triptans, ergots) that inhibit the trigeminal release of the above neuropeptides, block both the NV and PPE components of NI; nonetheless, it is not clear the extent to which their antimigraine efficacy arises from the inhibition of NV and PPE, or from inhibiting only one of these two mechanisms [121,122]. Interestingly: (i) the single use of olcegepant (a CGRP receptor antagonist), which only blocks CGRP-induced NV, showed acute antimigraine efficacy; and (ii) many drugs that exclusively block PPE are devoid of antimigraine efficacy [121,122]. Accordingly, the NV component of NI, but not PPE, may play a key role in the pathophysiology of migraine.

Based on these seminal findings, other lines of evidence suggest that acute antimigraine drugs, such as sumatriptan, dihydroergotamine and rizatriptan can inhibit the cranial NV produced by trigeminal release of CGRP [123–127]. Likewise, and most importantly within the context of the present review, the prophylactic antimigraine drug BOTOX decreases interictal CGRP plasma levels in patients with chronic migraine [128]. In keeping with these findings, other lines of clinical and preclinical research have shown that BOTOX: (i) decreases plasma levels of CGRP in patients

with trigeminal neuralgia [129] or chronic migraine [128]; (ii) inhibits mechanical allodynia, inflammatory cell infiltration and the increase in CGRP levels in cranial dura (an indicator of trigeminal activation) in a rat model of trigeminal pain [130]; (iii) cleaves SNAP-25 (which colocalizes with CGRP), resulting in inhibition of CGRP release [130]; and (iv) is transported axonally (when injected peripherally) to the trigeminal ganglion and, by transcytosis, reaches the CGRP-containing dural nerve endings [130]. Hence, inhibition of cranial NV is also a plausible mechanism of action of prophylactic antimigraine medications.

3.4. An increase in cranial vascular tone and other vascular actions

As previously pointed out, if dilatation and inflammation of cranial (i.e. extracranial and intracranial) arteries may result in migraine headache, prophylactic antimigraine medications may exert cranial vascular actions by increasing the vascular tone (producing vasoconstriction), blocking the vasodilatation induced by mediators of neurogenic inflammation (including CGRP), and inhibiting the inflammation of extracranial and intracranial arteries [11,19,101]. For example, in a migraine model in anaesthetized pigs, i.v. administration of olcegepant (a CGRP receptor antagonist previously known as BIBN4096BS) produced no effects on mean arterial blood pressure, but it was capable of dose-dependently blocking the carotid (i.e. cranial) vasodilator responses produced by intracarotid artery infusions of either: (i) capsaicin, which produces endogenous (trigeminal) release of CGRP [131]; or (ii) exogenous α -CGRP [131]. These findings demonstrate that olcegepant, as an antimigraine agent, antagonizes the cranial vasodilator responses induced by endogenous release of CGRP [131] or exogenous α -CGRP [131] by peripheral mechanisms.

Other cranial neurovascular mechanisms that may be relevant in migraine prophylaxis are those related with pharmacological actions in the CNS. Indeed, central mechanisms may also inhibit the cranial vasodilatation produced by endogenous (trigeminal) release of CGRP. For example, in a migraine model in anaesthetized dogs, the external carotid (cranial) vasodilatation produced by intracarotid artery infusions of capsaicin is dose-dependently inhibited by intrathecal (spinal C₁–C₃) administration of: (i) the selective α_2 -adrenoceptor agonist B-HT 933, with activation of spinal $\alpha_{2A/2C}$ -adrenoceptors [132]; (ii) the antimigraine agent sumatriptan, with activation of spinal 5-HT_{1B} receptors [133]; and (iii) the antimigraine agent dihydroergotamine, with activation of spinal 5-HT_{1B} and $\alpha_{2A/2C}$ -adrenoceptors [127]. Likewise, this cranial vasodilator response to capsaicin is also inhibited by oral chronic treatment with the selective serotonin reuptake inhibitor fluoxetine [134]. Taken together, these findings suggest that cranial neurovascular mechanisms that may be relevant in migraine prophylaxis are exerted at the peripheral and central levels. Depending on the specific prophylactic agent, each these four mechanisms may be induced to a varying degree (see below).

3.5. Specific mechanisms of the recommended prophylactic antimigraine medications

3.5.1. Antiepileptics: topiramate, divalproex sodium and sodium valproate

Several trials have demonstrated that topiramate, divalproex and sodium valproate (which can cross the BBB) are clinically effective for preventing migraine attacks [135–139]. Multiple ionic channel targets have been identified for topiramate's actions including, among others: phosphorylation-mediated inhibition of voltage-gated Na^+ channels [139,140]; blockade of L- and R-type Ca^{2+} channels [141,142]; and modulation of voltage-gated K^+ channels [143]. Moreover, inhibition of glutamate-mediated excitatory neurotransmission through the antagonism of AMPA and kainate receptors, facilitation of type A GABA receptors, and inhibition of carbonic anhydrase activity are some of the possible antimigraine mechanisms of topiramate [144]. Alternatively, topiramate may also attenuate neurogenic dural vasodilatation, presumably by a prejunctional inhibition of the release of CGRP from perivascular trigeminal neurons [145].

Furthermore, divalproex sodium is a mixture of equal quantities of sodium valproate and valproic acid. Like topiramate, it exerts its antimigraine actions by various mechanisms, including: (i) suppression of neuronal high-frequency firing via blockade of voltage-gated Na^+ [146] and low-threshold T-type Ca^{2+} channels [147]; (ii) increase in GABA inhibitory activity by promoting its availability via activation of glutamic acid decarboxylase (GAD; a GABA synthesizing enzyme); (iii) inhibition of GABA aminotransferase (GAT), part of the enzymatic degradation process of GABA; and (iv) enhancement of the responses produced by stimulation of both GABA_A and GABA_B receptors. The actions of divalproex sodium on the GABAergic system probably confer it its ability to reduce the neuronal activity of the cortex, perivascular parasympathetic nerves, and trigeminal nucleus caudalis [137,148].

The prophylactic actions of divalproex sodium may also be due to the reduction of brain aspartate levels and the suppression of NMDA-evoked transient depolarizations in neocortical pyramidal cells attenuating cortical activity [148]. Likewise, valproic acid can also induce peripheral inhibition of elevated CGRP immunoreactivity in trigeminal ganglia, implying that it reduces neurogenic inflammation and its nociceptive transmission [149].

3.5.2. β -blockers: propranolol, metoprolol, timolol and atenolol

β -blockers (i.e. β -adrenoceptor antagonists) constitute a pharmacological group widely used in migraine prevention, and multiple clinical trials support their effectiveness [150,151]. Transductionally, β -adrenoceptors are G_s -coupled metabotropic receptors that increase cyclic adenosine monophosphate (cAMP) and exert a local regulation of blood vessels during exercise and situations that increase catecholamines via the β_1 and β_2 -adrenoceptor types [152]; whereas β_3 -adrenoceptors seem to be upregulated during vascular abnormalities such as hemangiomas [153,154]. Certainly, most β -blockers effective in migraine prevention, including propranolol, metoprolol and timolol penetrate well

into the CNS [155]. In contrast, the water-soluble β -blocker atenolol does not cross the BBB [155], but it is still useful in reducing migraine activity [156]. Admittedly, it is still unclear: (i) if penetration into the CNS is a *conditio sine qua non* for the efficacy of β -blockers in the prevention of migraine; or (ii) whether atenolol has a different mechanism of action than lipophilic β -blockers, whose preventive actions could be exerted mainly centrally by modifying neuronal excitability [157].

Interestingly, propranolol and timolol interact with the serotonergic system by blocking 5-HT_{2B} and 5-HT_{2C} receptors [158,159]. Since 5-HT has been involved in migraine pathophysiology [11], it is suggested that antagonism on these receptors would lead to migraine prevention. Moreover, β -blockers: (i) inhibit the release of glutamate in the rostral ventrolateral medulla [160] as well as the β -adrenoceptor-mediated potentiation of glutamatergic release (e.g. in the striatum) [161]; (ii) exert synergistic effects with NMDA blockers, which reduce neuronal activity [162]; and (iii) may inhibit nitric oxide production by blocking iNOS, an effect mediated by β_2 -adrenoceptors [115]. Indeed, nitric oxide plays a role in primary headaches (including migraine) by activating the trigeminovascular complex [163], suggesting a possible mechanism for migraine prevention [164].

Furthermore, as part of its mechanisms for migraine prevention, propranolol can block/inhibit: (i) trigeminal nociception through β_1 -adrenoceptor blockade on ventroposteromedial thalamic neurons [165]; (ii) the chronic sensitization of pain descending control arising from the rostral ventromedial medulla and locus coeruleus [116]; (iii) the facilitation of trigeminovascular transmission within the trigeminocervical complex [116]; and (iv) the spontaneous firing of dural trigeminovascular neurons [166]. Although β -blockers' effects are exerted mainly centrally, their antimigraine actions may also involve peripheral blockage of pain promoter transmitters (e.g. nitric oxide, bradykinins, etc.) [167]. Indeed, β_1 -adrenoceptor blockade attenuates adrenaline and noradrenaline effects, thereby inhibiting the sympathetic nervous system's stimulating response [168]. Besides, this blockage results in: (i) a reduction in tyrosine hydroxylase activity in the superior cervical ganglia, which is a rate-limiting step in noradrenaline synthesis [169]; and (ii) a delayed reduction of the locus coeruleus neuron firing rate [170]. Finally, a cocktail of adrenoceptor blockers [i.e. propranolol (10 mg/kg) + prazosin (10 mg/kg) + atipamezole (1 mg/kg)] facilitated K^+ homeostasis after its alteration induced by CSD in mice [171].

More recently, an ophthalmic solution of timolol was discarded against migraine in a randomized, double-blinded, placebo-controlled, cross-over study [172]. Its penetration into the CNS is limited by its hydrophilic profile [173]; accordingly, its antimigraine actions may be mainly mediated by peripheral mechanisms, which remain to be determined.

3.5.3. Antidepressants: venlafaxine and amitriptyline

Amitriptyline and venlafaxine (serotonin-noradrenaline reuptake inhibitors, SNRIs) have been incorporated into migraine's preventive treatment guidelines. Current lines of evidence have shown the efficacy of amitriptyline (a tricyclic antidepressant), while few clinical studies have analyzed the preventive

efficacy of venlafaxine [150]. However, there are no significant differences in the efficacy between amitriptyline and venlafaxine [174].

As a SNRI, amitriptyline: (i) enhanced the actions of serotonin and noradrenaline neurotransmission in endogenous analgesic mechanisms via descending inhibitory nociceptive pathways from the brainstem to the trigeminal nucleus caudalis and spinal cord [175]; (ii) reduced CSD propagation [103], probably through inhibition of Na_V channels [167], downregulation of 5-HT_2 receptors, enhancement of GABA-mediated inhibition (by positive modulation of GABA_A receptors), and inhibition of the GABA transporter types 1 and 3 [162]; and (iii) produced a local peripheral antinociceptive action by inhibition of adenosine uptake and activation of adenosine A_1 receptors on sensory terminals [176].

Similarly, venlafaxine (unrelated to tricyclic antidepressants) induces migraine prophylaxis by inhibition of serotonin and noradrenaline reuptake at the presynaptic level [177]. Indeed, some studies have shown that venlafaxine: (i) inhibits serotonin reuptake at a dose less than 100 mg/day, whereas that of noradrenaline reuptake required a dose range of 100 to 375 mg/day; (ii) is a 30-fold more potent inhibitor of serotonin reuptake than noradrenaline reuptake [178]; (iii) also inhibits the reuptake of a small amount of dopamine at high doses [179]; and (iv) displays a very low affinity for adrenergic, muscarinic and GABAergic receptors, while it fails to inhibit Na_V channels [180].

3.5.4. Longer-acting triptans: frovatriptan, naratriptan and zolmitriptan

Overall, triptans are acute antimigraine molecules whose misuse can lead to medication overuse headache (MOH) [12]. However, the longer-acting triptans (i.e. frovatriptan, naratriptan and zolmitriptan) are also beneficial in the short-term prophylaxis of menstrual migraine [181,182]; this should not exceed 2–3 days a week (as recommended for other abortive medications) to avoid transformed migraine and MOH [183,184]. On the other hand, a prolonged use of triptans prophylaxis could be related with higher CGRP levels, worsened migraine attacks and/or diverse CNS disorders (e.g. neurotoxicity, behavioral alterations, etc.) [185,186]. Therefore, long-term use of triptans in pure menstrual migraine should be avoided or carried out under a strict medical follow-up.

From a pharmacological perspective, the triptans are agonists at $5\text{-HT}_{1B/1D/1F}$ receptors [11]; accordingly, their prophylactic antimigraine actions may be due to activation of these receptors. Indeed, the triptans: (i) block peripherally the transmission of pain signals from the dura to central trigeminovascular neurons [187], by presynaptic $5\text{-HT}_{1B/1D/1F}$ receptors in the dorsal horn to block synaptic transmission between the peripheral trigeminovascular neurons and cell bodies of their counterparts [188]; and (ii) activate 5-HT_{1B} receptors in the meningeal and extracerebral arteries, which induces a cranial vasoconstriction that would prevent the induction of migraine [189]. Notwithstanding, activation of $5\text{-HT}_{1B/1D/1F}$ receptors also inhibits the release of CGRP and substance P from trigeminal sensory fibers which, in turn, would inhibit: (i) trigeminally-mediated vasodilatation of dural

meningeal arteries; and (ii) dural-evoked central trigeminovascular responses [123,126,190–192].

4. The novel specific anti-CGRPergic prophylactic antimigraine medications

4.1. An overview on vascular CGRPergic transmission and its role in migraine pathophysiology

CGRP can be found in the central and peripheral nervous systems and, since its discovery, it has been mainly related with vasodilator actions [193,194]. At the peripheral level, CGRP is located in sensory nociceptive C- and A δ -fibers innervating several vascular beds [195]. When released from these perivascular sensory fibers, CGRP induces a potent vasodilation mediated by the CGRP receptor located in vascular smooth muscle and endothelium [196]. Regarding its structure and transduction mechanisms, this receptor is an atypical G-protein coupled receptor (GPCR); that is, to be functional, it requires heterodimerization with a receptor activity-modifying protein type 1 (RAMP1) [197,198]. In general, activation of the CGRP receptor elicits an increase in cAMP mediated by activation of $G_s\alpha$ proteins. Since this neuropeptide is also located in sensory nociceptive fibers, its role in nociception is undeniable. Indeed, at the spinal level (also at trigeminal nucleus caudalis, Sp5c), the CGRPergic transmission facilitates the nociceptive input (for refs. see [197]).

It is noteworthy that CGRP can increase blood flow by decreasing vascular resistance [199], but intravenous (i.v.) olcegepant (a CGRP receptor antagonist) in healthy rats produced no change in mean blood pressure [200]. This finding suggests no functional relevance of CGRP in cardiovascular homeostasis in healthy conditions [200]. Likewise, in humans under healthy [201,202] or pathological (coronary artery disease) [203] conditions, CGRP receptor blockade (with olcegepant and telcegepant, respectively) produced no effect on blood pressure. This may imply that endogenous CGRP plays no role in blood pressure regulation, but the functional relevance of this neuropeptide can be unmasked under continuous blockade of CGRPergic function [204].

For example, CGRP may act as a buffer during ischemia favoring vasodilation and improving myocardial contractility [204]; the fact that acute CGRP receptor blockade does not affect blood pressure regulation may only suggest that the cardiovascular compensatory mechanisms (e.g. cardiovascular reflexes) are robust enough to maintain the cardiovascular set point. Consistent with this view, using the pithed rat model (in which the CNS is destroyed and no cardiovascular reflexes are operative), Avilés-Rosas et al. [205] showed that i.v. olcegepant could: (i) block the systemic vasodilator (vasodepressor) responses by stimulation of the sensory perivascular CGRPergic outflow or i.v. injections of exogenous CGRP; and (ii) enhance the systemic vasoconstrictor (vasopressor) responses by electrical sympathetic stimulation or i.v. injections of exogenous noradrenaline [205]. Indeed, current data support the notion that CGRP may play a cardioprotective role [206].

Consequently, the main concerns when using anti-CGRPergic molecules to treat migraine are based on potential cardiovascular issues [12,17,207,208]. Indeed, CGRP is linked to the pathophysiology of migraine as a key neuromodulator in the trigeminovascular system [18]. Briefly, during a migraine attack, an increase in CGRP can be found in the cranial venous circulation [209], whereas an i.v. infusion of exogenous CGRP elicits a migraine-like attack in migraine sufferers [210]. In keeping with these findings, specific antimigraine therapy (acute and prophylactic) directly or indirectly inhibits the CGRPergic transmission at the peripheral and/or central levels [211]. For example, apart from their direct vasoconstrictor effects, ergots and triptans inhibit presynaptically (at Sp5c) or prejunctionally (at the perivascular level) the release of CGRP by activation of GPCRs coupled to G_i proteins (mainly 5-HT_{1B/1D/(1F)} receptors and, in the case of the ergots, also $\alpha_{2A/2C}$ -adrenoceptors). Hence, one can infer that not only do these drugs enhance the vasoconstrictor tone, but they also limit the pain transmission and the CGRP-induced vasodilation [211,212].

In view of the role of CGRP in migraine, specific antimigraine pharmacotherapies were fostered, leading to the development of small-molecule antagonists for the CGRP receptor, namely, the gepants [14]. Currently, ubrogepant [213], rimegepant [214] and atogepant [215] have been approved in the USA by the Food and Drug Administration (FDA). Although ubrogepant and rimegepant are indicated for acute abortive therapy, atogepant is approved as a prophylactic drug, and rimegepant is under clinical trials to test its efficacy as a prophylactic treatment (Clinical Trial Identifier: NCT03732638).

Furthermore, in the last three years, four CGRPergic monoclonal antibodies (mAbs) have emerged; these new biomolecules not only are the first of their class to target a GPCR, but in fact are also the first specific prophylactic antimigraine treatments [14].

4.2. The case of atogepant

Initially, seminal clinical findings using olcegepant (BIBN4096BS; the first gepant) demonstrated the therapeutic utility of blocking the CGRP receptor [201], although it was only effective when given i.v. [18]. This motivated an intense scientific search that resulted in the advent of eight gepants, but pharmacokinetics limitations and hepatotoxicity concerns emerged and delayed the pharmaceutical development of these molecules, particularly for the first-generation gepants (for refs. see [15,209]). Fortunately, further research resulted in the development of three new gepants (second-generation gepants), which are currently in the clinical field, and only atogepant has been approved for its use in the prophylaxis of episodic and chronic migraine [215].

Although all gepants block the CGRPergic transmission, none of these molecules seems to produce cardiovascular side effects when given acutely [200,216]. For example, a supratherapeutic dosage of telcagepant (a first-generation gepant) in healthy volunteers did not affect vascular tone in peripheral and central vascular beds [217]. Nevertheless, as pointed out above, CGRP may play a protective role during pathological conditions. Within this context: (i) Mulder et al.

[218] have shown in a surrogate model of stroke in mice that olcegepant and rimegepant worsened the ischemic outcome by occlusion of the middle cerebral artery [218]; and (ii) Rubio-Beltrán et al. [219] demonstrated in vitro that atogepant and ubrogepant blocked the CGRP-induced vasorelaxation of human intracranial arteries without vasoconstrictor effects on human coronary arteries. These data imply that: (i) during a migraine attack, where CGRP is released from perivascular trigeminal fibers, gepants blocked CGRP-induced vasodilation; and (ii) these compounds produced no effects in the coronary circulation (at least during acute exposure). Hence, long-term safety studies with atogepant will shed light on the relevance of perivascular CGRPergic transmission.

Finally, a critical aspect about the neural effects of gepants lies mainly in the notion that the doses used do not reach relevant concentrations in the CNS. This would restrict the effects of these compounds to the periphery, as previously observed with telcagepant [79]. Still, second-generation gepants are structurally distinct and, consequently, further studies will be required to evaluate if these molecules are capable of crossing the BBB.

4.3. Monoclonal antibodies (mAbs) against the CGRPergic system

In 2007, Juhl et al. [220] showed that mAbs could be used to treat migraine, on the basis that an i.v. infusion of a mAb against CGRP successfully inhibited the CGRP-induced vasodilation in a migraine rat model. Certainly, the mAbs have entered the clinical arena to treat migraine in the last three years and were designed as prophylactic antimigraine treatments. One of the striking differences with other antimigraine drugs is that these compounds are proteins (immunoglobulins) with high molecular weight (≈ 150 kDa), a fact that limits their therapeutic action to the periphery [12,15,17,204]. Among the peripheral potential sites of action for the antimigraine mAbs, the trigeminal ganglion may be important because of its adjacent connection between the C-fibers (CGRP containing neurons) and the A δ -fibers, which express CLR/RAMP1 (CGRP receptor elements) [221]. Certainly, mAbs are highly specific, have long terminal half-lives, and do not raise concerns about hepatotoxicity [220].

The antimigraine pharmacotherapy with CGRPergic mAbs can be divided into: (i) mAbs against the CGRP receptor (i.e. erenumab [222]); and (ii) mAbs against CGRP itself, that is, CGRP scavengers (galcanezumab [223], fremanezumab [224] and eptinezumab [225]). Furthermore, these human(ized) mAbs: (i) induce fewer or no side effects as compared to the established acute and nonspecific prophylactic drugs [12,17]; and (ii) decrease the monthly migraine days in patients with episodic or chronic migraine [226,227].

Regarding the interactions with the neurovascular system, although the above mAbs interfere with the CGRPergic system, the way this interaction occurs (blocking the receptor or acting as CGRP scavengers) exerts a differential effect on the intracellular events related to CGRP receptor activation. For example, erenumab, but not fremanezumab, induced CGRP receptor internalization [228], meaning that apart from the antagonist action of erenumab, it could also limit the presence

of membrane receptors. In contrast, the fact that mAbs against CGRP limit the circulating levels of CGRP could probably lead to an overexpression of CGRP receptors, a hypothesis that needs to be further investigated. In any case, the therapy with CGRPergic mAbs is considered effective and safe [229].

5. Conclusion

As a whole, the medications effective in migraine prophylaxis probably involve the predominant induction of neurovascular mechanisms that may include, among others: (i) a reduction in the CSD-associated events; (ii) an inhibition of central and peripheral pain sensitization; (iii) the blockade of neurogenic inflammation; and/or (iv) an increase in cranial vascular tone and other vascular actions. Admittedly: (i) each of these four mechanisms may be induced to a greater or lesser extent depending on the prophylactic agent in question; and (ii) there may be more mechanisms involved, but this is a relatively new research area whose future advances will shed further light on this possibility.

6. Expert opinion

In general, prophylactic (preventive) antimigraine treatment focuses mainly on: (i) the reduction of attack's frequency and/or severity; and (ii) the improvement of drug responsiveness to acute attacks [13]. However, most nonspecific prophylactic antimigraine therapies are sometimes unsatisfactory due to, *inter alia*, adverse side-effects, restricted efficacy and drug interactions [230]. Furthermore, until recently, the mechanisms of action of most prophylactic antimigraine drugs had remained typically unknown or unproven, but some lines of evidence [100,101] support at least four common pathways, namely: (i) a reduction in the CSD-associated events; (ii) an inhibition of central and peripheral sensitization of pain pathways; (iii) a blockade of neurogenic inflammation; and (iv) an increase in cranial vascular tone and other vascular actions [11,19,101].

As with acute (abortive) medications, the selection of a given prophylactic medication must consider, among others, patient's age, comorbid conditions and the side-effect profile [231]. Within this context, antiepileptics, antidepressants and antihypertensives may be considered as an initial prophylactic treatment since they are more affordable. For any of these initial prophylactic agents, prophylaxis should not be considered a failure until it has been given at the maximum tolerable dose for at least 30 days [231]. In recent times, the prophylactic antimigraine treatment (previously consisting mainly of nonspecific drugs during decades; see above) has witnessed a significant progress represented by the development of specific prophylactic agents. In fact, the FDA has approved one gepant (*i.e.* atogepant in 2021) and four mAbs (*i.e.* erenumab, galcanezumab and fremanezumab in 2018, as well as eptinezumab in 2021) for this purpose [15]. In view of their interference with the cranial neurovascular CGRPergic transmission, these novel prophylactics can certainly reduce the endogenous CGRP-induced vasodilation of dural arteries.

Although these molecules are effective in migraine prophylaxis in general, some safety questions should be answered in the coming years with the extensive and long-term use of

these compounds. As recently reviewed by Rivera-Mancilla et al. [17], these agents are safe in the short-term, but their long-term effects and safety have not yet been explored. Hence, long-term phase IV clinical studies (including patients with cardiovascular comorbidities) will shed further light on the safety of these novel prophylactics. Indeed, a recent 5-year open-label extension study using erenumab suggests that this mAb does not impact the cardiovascular system [232]. However, the long-term treatment with these mAbs, even if devoid of cardiovascular side effects, could favor the induction of autoantibodies, a fact that would represent a major adverse event when using these novel prophylactics [15].

In view of the peripheral actions of these mAbs at the cranial neurovascular level, perhaps future research strategies could contemplate the use of nanotechnology for the development of more effective prophylactic delivery systems that can cross the BBB and transport these mAbs to CNS areas related with the modulation of pain transmission [15].

On the other hand, the fact that specific prophylactic antimigraine drugs seem not to be effective in all migraine patients implies that mediators other than CGRP could be involved. In support of this notion: (i) mAbs against the pituitary adenylate cyclase-activating peptide (PACAP) system are in the pipeline as potential antimigraine drugs [14]; (ii) the use of specific CGRP neutralizing L-aptamers also has a potential in migraine therapeutics, and due to their high molecular weight they do not cross the BBB [233]; and (iii) oxytocin (a hypothalamic neuropeptide) has been used as an analgesic, and it shows antimigraine action when given intranasally [14]. Although their neurovascular actions have not yet been documented when used in antimigraine regimens, these findings suggest that a multi-target drug approach might represent a future development for effective antimigraine therapies [15]. Finally, it would be interesting to investigate these novel prophylactic agents in patients suffering from treatment-resistant migraine.

What is unquestionable, thus far, is that the novel specific prophylactic treatments: (i) can induce neurovascular effects by interfering with the cranial vasodilator CGRPergic transmission; and (ii) promise to be more effective, devoid of significant adverse effects (unlike nonspecific treatments), and more beneficial to the quality of life of migraine patients.

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Figure 1 was drawn using biorender.com with license paid by the Autonomous University of Aguascalientes to BAMC.

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